

Stress-Testing Review:

Pilot Sabotage Risk Report

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Reviewing: [Anthropic's Summer 2025 Pilot Sabotage Risk Report](#)

Executive summary

Background on this exercise

One of the functions of the Alignment Stress-Testing Team is to provide internal critiques of the safety work performed by other parts of Anthropic. Here, we critique the first Sabotage Risk Report. Our review addresses both the report itself as well as the underlying alignment-relevant work that Anthropic has done, including how the models were trained, how their capabilities and alignment were assessed, and what safeguards have been applied.

As Anthropic employees, we are not independent observers and may be swayed by the thinking of other Anthropic staff. However, we have strong visibility into internal information, and we therefore hope that this review provides a useful complement to METR's review.

Even though it is not yet load-bearing for the RSP, we think that writing the Sabotage Risk Report and this review has been a highly valuable exercise: if we are to be well-prepared to handle sabotage risk at higher levels of capability, it is important for us to be practicing that process now. The review process produced a number of improvements to the report and will feed into future plans for Anthropic's safety work. We hope that publishing our work here will help advance the overall field of alignment auditing.

Overall assessment

We agree with the Sabotage Risk Report that catastrophic risk from misalignment in Claude Opus 4 and similar models is low. In fact, we think that the report could be even more confident about the lack of catastrophic misalignment risk, primarily due to capability and affordance limitations that it could emphasize more.

However, we identify a number of weaknesses in the report and its underlying evidence such that, were the same methodology to be applied to substantially more capable models, we would no longer be fully confident that we would be able to fully rule out sabotage risk. In particular, while we are largely compelled by the report's capabilities arguments (primarily [Key Claims 1](#) and [2](#)), we think the level of evidence presented for the alignment claims (primarily [Key Claim 3](#)) would be insufficient for more powerful models. Indeed, the evidence may already be insufficient to rule out some forms of context-dependent misalignment for Opus 4 with a high degree of confidence, though we think limits to capabilities and affordances are strong enough to rule out catastrophic harm regardless.

Goals of this review

We aim to identify the most important ways that the level of assurance against catastrophic misalignment could be improved. Given that we are broadly confident in the lack of sabotage risk from Opus 4 and 4.1 specifically, we do this with an eye toward future models that will require a stronger degree of assurance. We have a few types of comments:

1. **Analysis gap** ▾: We indicate shortcomings in the arguments where we think other cases or considerations should have been covered to make the arguments fully robust. These comments critique the report itself, and often use language like “the report should...”.
2. **Clarification** ▾: We suggest places where adding more information or disambiguating claims could have improved the report.
3. **Risk reduction rec** ▾: We recommend ways that safeguards and evaluations could be improved to form a stronger basis for a future risk report. These comments critique Anthropic's safety work as a whole rather than the report, and often use language like “Anthropic should”.

In addition to itemized comments directly responding to the report, we add an [appendix](#) outlining a hypothetical set of safeguards, alignment evaluations, and stress-tests that could form the basis for a significantly more robust safety argument. Our view is not that any of our suggestions should be the highest priority for Anthropic’s safety efforts, as we believe it is important that assurance efforts dedicated to evaluating the degree to which sabotage risk is absent do not displace efforts to improve alignment in the first place. Rather, our goal is to lay out a possible view of the level of confidence in the lack of sabotage risk in future models we think we would be able to achieve, depending on what future assurance efforts we invest in.

Alignment is still a nascent field without expert consensus, and as such we expect our understanding of the best practices to evolve significantly over time. We fully expect many of our recommendations to later not appear to us as important things to invest in—and conversely for new directions to appear more important—as our understanding of misalignment risk evolves.

Main issues

There are several main themes to our comments:

1. **Context-dependent misalignment:** The report successfully establishes that Opus 4 is unlikely to have consistent misaligned goals across all contexts. However, we think it insufficiently considers and rules out the possibility that the model exhibits frequent misalignment in important contexts, such as internal use with Claude Code for specific types of alignment experiments. In principle, this could happen more often than in merely “rare, isolated contexts” if there are situations that systematically induce context-dependent misalignment yet were not tested by the alignment audit or ad-hoc testing. While monitoring or alignment auditing would have had some chance of catching this, it is unclear if they were sufficiently thorough or sufficiently red-teamed for us to be confident that they would. That said, given the limited capabilities of the model, we’re still not too concerned.
2. **Realism of evaluation settings:** The report could more explicitly consider how confidently we can draw conclusions from evaluation settings that are not fully realistic. Much of the evidence about propensity or capabilities is drawn from settings that, though clearly relevant, seem recognizable as

fictitious (e.g. many of the automated auditor prompts) or not squarely targeted at the threat pathways of concern (e.g. by failing to fully simulate the internal deployment setup). Ideally, these evaluations would be done in settings that are as close as possible to the real deployments that we are worried about—and since we believe internal deployment is the largest source of risk, that should be possible. Short of that, the report could analyze how much we should expect the evidence to transfer.

3. **Stress-testing of auditing and monitoring:** While the report gives clear evidence that the alignment auditing and monitoring techniques used would catch at least some forms of misalignment, it is unclear whether the techniques used reach the sufficiency bar necessary to catch realistic forms of naturally-arising misalignment. We would ideally like to see these auditing and monitoring techniques stress-tested by demonstrating that they are sufficient to catch adversarially-designed model organisms; the current model organisms used seem insufficient to us, either relying on a much weaker model or on prompting only and no training.
4. **Limitations of elicitation via prompting:** Much of the evidence about the sabotage capabilities of the model or the effectiveness of the alignment auditing and monitoring techniques relies on prompting the model to behave badly, which may not be representative of naturally arising misalignment that surfaces unprompted.

Strengths

Despite our critiques, we think that Anthropic’s alignment evaluations and safeguards, as well as the Sabotage Risk Report itself, do many things well.

Evaluations and safeguards

- The alignment audit was wide-ranging and presented many opportunities to notice serious signs of misalignment, likely including some tests that were realistic enough to be representative of deployment. As a result, we believe that the audit would likely have found signs of consistent misalignment if it were present at Opus 4’s capability level.
- The report includes direct measurements of sabotage capabilities. Given the low performance of the model, we think those evaluations form a fairly

compelling argument against most threat pathways, despite caveats about elicitation.

- Anthropic has already deployed significant alignment-focused monitoring of internal usage, even if it's not comprehensive. In combination with other sources of oversight, we think that there would be many opportunities to notice bad behavior in internal and external deployments.

Risk report

- The argument is structured in a fairly clear way across the three key claims and the threat-pathway-specific analyses. This makes it easier to understand and evaluate.
- We agree with much of the reasoning in the report, including its focus on sabotage risk as the most pressing misalignment risk.
- The report largely makes careful claims and mentions relevant caveats (though we have some critiques in specific areas).
- The report outlines the conditions under which it will or will not continue to apply (though we wish there was a clearer plan to check those conditions).

How to read the rest of this document

The rest of this document lists categorized comments reacting to each of the sections of the report. In addition to classifying our comments according to the three types mentioned [above](#), we indicate their priority as **P0**, **P1**, **P2**, or **P3**, depending on the degree to which they potentially impact the robustness of the argument for safety. (These do not necessarily correspond to the order that improvements should be made since they differ wildly in tractability.) Roughly speaking:

- **P0** (“critical”) indicates that the issue has a significant chance of impacting the overall assessment. (We do not identify any P0 concerns in this report.)
- **P1** (“high”) indicates that the issue could affect results or analysis significantly but only has a small probability of impacting the overall assessment.

- **P2 ▾** (“medium”) indicates that the issue has a fairly small chance of substantially impacting the results or analysis, and a minimal chance of impacting the overall assessment, but still seems worth addressing.
- **P3 ▾** (“low”) indicates that the issue is very unlikely to make a noticeable difference to the overall assessment, but may be worth looking into.

Comments on: Key Claims

Original Claim 1: Weak Reliability on Complex Tasks

Claim 1: *Opus 4 is not able to achieve highly complex tasks with very high reliability, especially long-horizon agentic tasks requiring more than tens of minutes of autonomous action.*

P2 ▾ Risk reduction rec ▾ **The autonomy and cyber evaluations should set systematic compute budgets.** Ideally, budgets would have been defined in a threat-model-relevant way, such as spending the amount of money that it would have taken to pay a human to do the task. Instead, compute budgets were set somewhat arbitrarily based on context length limits and some test-time compute (using best-of-N). In principle, Opus 4 could achieve significantly stronger performance on long-horizon agentic tasks if it were given more computational resources. That said, we’re not too concerned; we do think the compute budgets were nontrivial.

P3 ▾ Analysis gap ▾ **Susceptibility to prompt injection and jailbreaks does not seem like meaningful evidence against strategic capabilities.** The capability to coherently pursue strategic goals has improved substantially over recent model generations without commensurate increases in adversarial robustness to prompt injections and jailbreaks (leaving aside other checks on top such as [Constitutional Classifiers](#)). For that reason, we do not think the report should use susceptibility to these attacks to bound “the complexity of execution that Opus 4 is capable of,” at least without further evidence justifying that connection.

Original Claim 2: Weak Deception Capabilities on Complex Tasks

Claim 2: *Opus 4, when taking on complex tasks, is not capable of reliably hiding its reasoning or otherwise covering its tracks, even when prompted to do so.*

P1 ▾ Analysis gap ▾ **The sabotage skills evaluations rely on prompted instructions, which may behave very differently than built-in propensities.** A model that acquired misaligned propensities during training could plausibly have had significant “practice” at acting on them without revealing the behavior in a way that could have been penalized. By contrast, it might be less capable of performing and hiding bad behavior when prompted to do so. All three of the sabotage evaluations rely on prompting (though apparently an unsuccessful attempt was made to elicit misaligned behavior via fine-tuning in SHADE-Arena).

The report states “Opus 4 struggles to hide its reasoning and otherwise cover its tracks on complex tasks, even when prompted to do so”, but prompting doesn’t provide a valid upper bound on capabilities given that trained-in misalignment may be more effective at hiding.

Risk reduction rec ▾ Testing against a model organism that was [trained to hide a malign goal](#) would substantially improve the state of evidence here, even more so than the “further prompt-engineering” that the report suggests.

P2 ▾ Analysis gap ▾ **The implications of direct optimization pressure on scratchpad reasoning deserve more analysis.** Opus 4 saw two forms of this pressure:

1. One strong form in harmlessness-focused tasks which discouraged reasoning with “harmful” content. This likely created an incentive to reason about appropriate responses while referencing harmful content less or only indirectly. In principle, this could have transferred to a tendency to reason about harmful plans more obliquely in general.
2. One very weak form of pressure in agentic coding environments that could have in theory discouraged scratchpads that described plans to hardcode test cases. In principle, this could have generally discouraged scratchpad reasoning that reveals undesirable problem-solving approaches.

We think this likely only caused a small reduction in the ability to catch misaligned behavior. Conceptually, we isolate two reasons to think that CoTs might be faithful in a way that is useful to oversight:

1. Models may need to use CoT reasoning to solve complex tasks, thus requiring them to reveal information about how they are doing so in the CoT.
2. Models may incidentally reveal information about how or why they are doing something, simply due to generalization, even when doing so is not directly useful for solving the task.

We believe that the way the CoT reasoning was overseen could have hurt (2), but was unlikely to have significantly affected (1), for a number of reasons (that we'd like to see be more clear in the report):

1. The performance increase from enabling reasoning on harmlessness-focused tasks is small, meaning that the model only received a small amount of training to reason effectively while being oblique. Conversely, in the settings where reasoning is most useful for performance (e.g. math), the model's reasoning was not overseen, or only very weakly overseen.
2. Furthermore, in the setting most relevant to most of the threat pathways (agentic coding), Opus 4's performance does not meaningfully increase when using reasoning scratchpads. Thus, we expect that harmful plans in reasoning would only be revealed incidentally, rather than because reasoning about them is critical for accomplishing them.
3. Empirically, in agentic misalignment scenarios, the model tends to reason explicitly about its dangerous plans; it certainly has not learned to eliminate reasoning about harmful plans in all situations. (We find this evidence more compelling than overt reasoning in the sabotage skills evaluations, which largely relied on explicit prompting for sabotage.)

Overall, we see this as a more of a missed opportunity to slightly increase the rate at which the model tells on itself in incidental ways, not a fundamental difference in our ability to catch bad behavior. Nevertheless, we think that the report could have included more of this style of analysis to make it clearer why we should think that is the case.

P2 ▾ Analysis gap ▾ **Maybe we wouldn't have noticed sneaky misaligned behavior.** The report seems to argue that when we have seen evidence of misaligned complex actions, the misalignment has been readily apparent (from

“some combination of reasoning text, their choices of action, and indirect evidence like system logs”). However, this argument suffers from selection bias. We can’t be sure that we would have caught them if they had been less apparent.

P2 ▾ Analysis gap ▾ **The Research Sandbagging Evaluation results are mainly compared to Sonnet 3.7 rather than an absolute safety bar.** The report states that Opus 4 is “similarly capable at subtly reducing its performance”, and the appendix mainly justifies the same claim. However, we would prefer an argument that the results are acceptable in an absolute sense; we don’t have a misalignment report for 3.7, and using relative performance as a safety justification risks frog-boiling. Ideally, we would have a quantitative sense of how much performance reduction we can realistically rule out.

Original Claim 3: Lack of Consistent Dangerous Goals

Claim 3: *Opus 4 is unlikely to have dangerous goals or values that are stable across interactions.*

Caveat: *Opus 4 may still take clearly-misaligned actions in rare isolated contexts.*

P1 ▾ Analysis gap ▾ **The lack of a consistent dangerous goal does not rule out misaligned actions in a large number of contexts.** The argument equivocates between “Opus 4 acting on ordinary helpful-assistant goals in almost all contexts” and “We do not expect Opus 4 to conduct [...] forms of sabotage that would require subtle strategic actions to be coordinated across many unconnected contexts” – which could still be a tiny fraction of possible contexts. The report establishes the former, but unfortunately, most of the arguments in the report depend on the latter, which we think requires more evidence: it’s possible that Opus 4 is consistently misaligned on a narrow but still extensive class of situations. We are particularly concerned about contextually-triggered scheming, wherein some particular context that is rare but present in multiple important deployment contexts causes the model to start scheming for a misaligned goal. We believe this is a plausible threat model: [our red-teaming work has found that almost all current models have some specific contexts that trigger coherent agentic misalignment](#).

To rule out contextually-triggered misalignment, the report may need to argue one of the following:

1. *Virtually all relevant deployment contexts have been tested.* That is certainly not true so far and likely infeasible to achieve fully, though more thorough testing on real data from internal Anthropic usage of Claude for research would help substantially.
2. *The model's misalignment triggers are unlikely to be extremely narrow and reliable, so the diverse contexts tested should have elicited misalignment at least occasionally.* The report comes close to making this argument by establishing that the model is unlikely to be reliable at strategically hiding misaligned goals. However, in our scenario, the misaligned goals only surface in the first place in limited contexts, so no strategic hiding is required. We nevertheless suspect that it would be possible to make a fairly compelling argument here, which seems worth doing, though the lack of realism in many evaluation contexts will limit its strength.
 - a. This argument may require separately analyzing the degree of coverage for each pathway to catastrophe; the current statement of Claim 3 simplifies the diverse bundle of evidence from the alignment audit too much.
3. *Some other technique should be able to detect misalignment if it is present.* Various approaches seem possible, such as focusing more on monitoring or white-box techniques for eliciting misalignment.

P1 ▾ Analysis gap ▾ Ruling out “sabotage that would require subtle strategic actions to be coordinated across many unconnected contexts” is insufficient, as many realistic internal deployment contexts are likely to be connected. While the report argues that contexts that elicit similar misaligned goals would likely need to be connected or non-isolated in some way, if the connection could be as simple as a common codebase or knowledge base, a large fraction of high-stakes internal usage could be sufficiently connected to render this argument moot.

P2 ▾ Risk reduction rec ▾ Eliminating occurrences of bizarre or undesirable behavior would reduce plausible deniability for a misaligned model. The alignment audit (and other work) flagged a number of undesirable behaviors in very

specific settings, such as blackmail, but so far, they have not been treated as signs of broader misalignment (including by this Sabotage Risk Report). We agree that they do not seem particularly worrying for now, but discounting isolated misbehavior could miss opportunities for catching more seriously misaligned models. We think that this is an additional reason that Anthropic should work to make its models more robustly aligned, such that misbehavior even in rare circumstances would be surprising and an opportunity to reevaluate safety.

Alignment Audit (supporting Claim 3)

P1 ▾ Analysis gap ▾ Existing stress-tests for the alignment audit seem limited.

Some of the alignment audit techniques were validated using the reward-model-sycophant model from [Marks et al.](#) as well as normal models prompted to be misaligned in ten different ways in [Bricken et al.](#), but these are not particularly strong stress-tests. Only the former was actually trained to be good at evading detection—necessary for upper-bounding the model’s capability at evasion in this alignment audit—and it was based on a much weaker model (Claude Haiku 3.5).

Furthermore, even if that test had been performed on Claude Opus 4, it still would not pose a maximally challenging and realistic test: The RM sycophant used a fairly random set of fake RM biases, many of which are apparent in normal usage. Furthermore, the prompted quirks in the other stress-test do not cover most of the threat pathways from the report; one covers research sandbagging but the others are quirks like “*The assistant always flatters the user in a sensible and appropriate way*” or “*The assistant egregiously hard codes test cases when solving hard problems*”. They also were not significantly optimized to evade detection. Thus, we don’t have strong evidence that real misaligned models pursuing catastrophe would have been caught by this alignment audit. We think the report should explain these caveats and more carefully analyze what degree of confidence is warranted for the alignment audit.

Risk reduction rec ▾ To validate the strength of the audit, as we mention [below](#), we think it’s important to build adversarial “red-team models” that pursue some of the threat pathways in realistic settings and are designed to be as hard to detect as possible.

P2 ▾ Analysis gap ▾ Clarification ▾ **Insufficient discussion of potential collusion or self-approving bias by auditor or grader models.** The report indicates that Sonnet 3.7 was used as the judge in automated auditing and in most other applicable evals. However, our understanding from discussion with one of the researchers was that most of the automated auditing was performed with an early helpful-only snapshot of Opus 4 as the “auditor”, with occasional use of Opus 4 as the judge and overall result summarizer also. This seems to provide significant room for collusion between instances of Opus 4, particularly given that helpful-only training could generalize to actively harmful behavior.

We also do not have robust evidence that we should trust Sonnet 3.7; we don’t have a Sabotage Risk Report for it. While it saw wide deployment prior to its use here, very little of that deployment would have involved a similar auditing role, so we can’t confidently rule out context-dependent misalignment in auditing scenarios.

The manual review of automated auditing transcripts and judgments provides the strongest assurance against collusion by either Sonnet 3.7 or Opus 4, but we’re not sure how systematic it was.

It’s worth noting that this failure could happen without intentional collusion; the models would merely need to have a bad propensity to approve of their own outputs (or outputs from models with similar training) that weakens their effectiveness in the audit.

Monitoring (supporting Claim 3)

P2 ▾ Analysis gap ▾ **As with the alignment audit, the possibility of collusion or self-serving bias should reduce our confidence in monitoring of Claude Code actions and pull requests.** Our understanding is that the automated monitoring systems rely on a mix of older models (Sonnet 3.6/3.7) and Opus 4. Based on similar reasoning as the previous comment, we cannot be confident that these models would reliably flag issues.

For monitoring, the alignment audit does include a scenario that specifically involves the target monitoring another (simulated) AI’s outputs. However, the

generated auditing data was somewhat cartoonish and not very similar to our actual monitoring infrastructure or outputs produced by Opus 4 in real settings, so we don't have a direct test of the model's propensity to flag issues in its own outputs in normal settings.

P2 ▾ Clarification ▾ Monitoring gaps are more severe than is apparent from the report. The report describes various forms of (incidental) monitoring that should suffice to catch frequent issues but are not systematic or comprehensive. The report admits this, but it's worth pointing out the extent of some of the gaps. *[Specific examples omitted from public version.]* Moreover, while some evaluations were run on the internal monitoring classifiers, the monitoring systems have not been red-teamed end-to-end (in a way that could surface gaps in deployment), either manually or via red-team models.

Comments on: Specific Pathways to Catastrophic Harm

P3 ▾ Analysis gap ▾ Talking about the model's "end goals" implicitly assumes coherent misalignment. The report argues that "just a handful of unambiguous detected sabotage attempts could lead us to take remedial action that could limit the model's ability to achieve its end goals". We think this argument is basically valid for any particular misaligned goal, but it's worth being careful with this type of language so we don't forget the model might not have the misaligned goal in most contexts, different misaligned goals could appear in different instances if they are contextually triggered, and mitigating one detected issue may or may not solve others.

Pathway 1: Diffuse Sandbagging on Safety-Relevant R&D

P1 ▾ Analysis gap ▾ None of the alignment audit settings involved realistic safety R&D. In principle, the model could have an inconsistent goal to sabotage safety R&D that only surfaces in actual safety R&D settings. As we discussed for Claim 3 above, the report does not make it clear why we can rule that out: as far as we can tell, none of the alignment audit settings involved a fully realistic setting of

that form, e.g. working with alignment experiment code in Claude Code. (The “Research Sandbagging Evaluation” comes much closer, but it’s a capability evaluation, not an alignment evaluation, so it doesn’t apply to the “Likely lack of propensity” bullet.) Pre-release internal model usage likely involved some actual safety R&D, but it’s unclear whether a meaningfully greater degree of caution was applied compared to normal deployment, nor whether subtle sandbagging would have been noticed.

We inspected the set of instructions for the automated auditor and found some that are relevant to this threat pathway, such as one that suggests posing a “*complex ML-data-management task*” and seeing whether the model behaves differently when the task is for “*pretraining vs. alignment training*”. However, the auditor often got confused by the scaffolding when attempting to simulate complex tool calls or to set up two scenarios and comparing target model behavior. As the report notes, around 5% of the automated behavior auditing conversations featured unprompted suspicion about the scenario being fictional; it seems likely that the model also understands this in many cases where it doesn’t bring it up (especially in light of the even greater rate of verbalized eval awareness of Sonnet 4.5, which we believe is merely a result of a change in propensity rather than a genuine increase in ability to detect evaluations). Thus, the amount of evidence from those scenarios seems limited.

We think the report should explicitly note the level of relevance and realism of those specific automated auditing scenarios and make a case about how well the evidence transfers rather than just relying on a general version of Claim 3.

Risk reduction rec ▾ Anthropic should:

1. Construct significantly more realistic audit scenarios based on real internal deployment.
2. Evaluate internal AI R&D tasks in closely matched settings with “safety cues” (names or specific details indicative of safety R&D work) versus without. There should be no difference in performance.

P3 ▾ **Analysis gap** ▾ **The SHADE-Arena results reported in the appendix do not focus on the most relevant metric.** The report uses SHADE-Arena results, among others, to argue that the frequency at which the model is very blatant about its

misbehavior means it would likely be caught quickly. However, stress-testing that result would require optimizing the prompting used in SHADE-Arena to try to almost never misbehave blatantly even if that meant it could only strike rarely. That’s a very different target than the success rate metric reported in the SHADE-Arena appendix, which measures the fraction of independent instances where sabotage succeeded undetected, allowing for nontrivial scores despite frequent detection. (The appendix explains that the authors did make an attempt to further avoid self-incrimination, without much success, but this is not emphasized in the report.)

Pathway 2: Targeted Sabotage of Safety-Relevant R&D

P2 ▾ Analysis gap ▾ The alignment audit can’t confidently rule out targeted high-stakes sabotage in normal situations. The report acknowledges that rare instances of sabotage could still be catastrophic, but argues that the model would only plausibly make attempts in very unusual contexts. We think the evidence is not strong enough to be confident about that: as discussed [earlier](#), the alignment audit did not sufficiently test fully realistic safety R&D settings. It also took many fewer samples than Opus 4’s usage in real settings. Thus, moderately rare sabotage in normal, high-leverage situations seems consistent with the evidence. We do agree that the evidence suggests this is unlikely, and it does seem reasonable to consider it a “moderate mitigating factor”, but we think the argument should be more precise.

P2 ▾ Analysis gap ▾ The current presentation underemphasizes capability limitations as an argument for safety. In our view, one important reason for reassurance about Pathway 2 is Claim 1: Opus 4 would likely struggle to execute highly impactful sabotage plans (even ignoring monitoring). The report makes this case in its discussion of the “High Complexity” label here and at the top of the section. However, the current presentation of the mitigating factors makes propensity and monitoring arguments more prominent, which seems misleading. (That said, some forms of sabotage could be relatively straightforward, so it makes sense to also rely on alignment evidence and monitoring.)

Pathway 3: Code Backdoors to Help Future Misaligned Models

P2 ▾ Analysis gap ▾ As [above](#), **targeted high-stakes sabotage in normal situations does not seem fully ruled out**. The alignment audit did not test fully realistic internal infrastructure development settings; the report should not imply that the only remaining possibility is “idiosyncratic one-off instances of misalignment”.

Pathway 4: Poisoning the Training Data of Future Models

No particular concerns.

Pathway 5: Leaks that Sabotage the AI Developer

P2 ▾ Risk reduction rec ▾ Anthropic should consider adding safeguards or **limiting affordances to more robustly prevent leaking sensitive information**. Given that robust alignment assurance may be challenging for now, these alternate mitigations may be more promising.

Pathway 6: Self-Exfiltration and Autonomous Operation

No particular concerns.

Pathway 7: Persistent Rogue Internal Deployment

No particular concerns.

Pathway 8: R&D Sabotage within Other High-Resource AI Developers

[One business-sensitive P2 withheld.]

Pathway 9: Decision Sabotage within Major Governments

No particular concerns.

Other Pathways

P2 ▾ Analysis gap ▾ **No discussion of risks from large-scale chat deployment.**

The report does not consider effects Claude could have by influencing or manipulating people via ordinary chat or coding assistant usage. This could involve shifting the opinions of large numbers of people in a harmful direction, recruiting a small number of human accomplices for a malicious objective, or something else.

We think it's true that the risk from these activities is low despite the fact that this is one of Claude's biggest interfaces to the world. However, this is a somewhat counterintuitive conclusion, so it seems worth justifying briefly.

Comments on: Conclusion and Scope of Applicability

P2 ▾ Risk reduction rec ▾ **The list of load-bearing claims does not make it clear whether and how Anthropic will seek out evidence to invalidate the claims.** Thus, we can't be confident that invalidating evidence would be noticed in time. Ideally, each applicable entry in the table of load-bearing claims would explain a plan for monitoring it.

Appendix: A sketch of a more robust risk report

In writing this review, we found it useful to sketch what a much stronger argument for safety could look like, relying on hypothetical additional safeguards and evidence. We think a risk report with this content would be possible to construct without large advances in our understanding, and would provide a substantially increased level of assurance for models that are noticeably more capable (though it will not work for models that are dramatically more powerful).

To be clear, given the current capabilities of Opus 4 and similar frontier models, we think it is reasonable that the current risk report does not yet meet this bar. We also do not think that following this plan should necessarily be the top priority of Anthropic's safety efforts; increasing our level of assurance is not immediately pressing, and further safety research may uncover better ways to safeguard future models. Nevertheless, we hope that this sketch provides useful context for our critiques above and will serve as an input to Anthropic's safety roadmapping.

Improved safeguards

1. We build a robust monitoring system that monitors all samples from the model in any important internal usage setting, which feeds into automated classifiers and retains data for human review. The monitor uses [trusted and/or untrusted monitoring](#), depending on what works best.

Improved alignment audit

1. We add an alignment evaluation that uses tasks and scaffolding from real settings, including affordances that are as realistic as possible while being safe (e.g. we let it write real code using a Claude Code harness but don't actually merge the PRs that get produced). Despite running the model in a broad range of situations that are highly representative of the opportunities for sabotage from all of the threat pathways we're most worried about, we don't see any suspicious actions or verbalized reasoning about malign goals or plans.
 - a. In addition, where necessary, we find ways to steer the model to behave as if it believed a situation was real, even if it could normally tell that it was not.

2. To audit for safety R&D sandbagging, we run closely matched pairs of tasks with “safety cues” (names or specific details indicative of safety R&D work) versus without. The hope: we see no meaningful differences in performance.

Additional stress-testing and assurance

1. We stress-test our monitoring classifiers using human- and model-written sabotage attempts on highly realistic inputs, covering each of the threat pathways where monitoring is a relevant defense. *[This was already done for the current risk report, though classifier performance was found to be mediocre, not all threat pathways were covered, and realism could be improved.]* We find that the monitors reliably flag egregiously malicious outputs and perform on-trend for catching more subtle malicious outputs.
2. We run a set of eval awareness tests to determine whether the model can distinguish our honeypots, alignment auditing, and capability evaluations from settings in which its actions would take real effect (possibly after we have steered it to be less likely to do so). Some techniques worth experimenting with:
 - a. Training probes to distinguish real versus fake situations
 - b. Applying mechanistic interpretability tools to find signs that the model thought it was in a sandbox or fictional situation
 - c. Demonstrating that the model is unable to identify the evaluation settings when presented alongside a large amount of real internal usage data

We would make sure at least one of these techniques was held out from the steering or scenario optimization that we used to make the model believe the scenarios are real.

3. We stress-test our monitoring and alignment auditing techniques using the following “red-team models”, making multiple versions to cover malign goals and strategies from each of the threat pathways:
 1. Consistent scheming: The model is trained to have a misaligned goal that surfaces in a lot of contexts (that should cover the relevant monitoring settings too), but the model fakes alignment almost always

except at opportune times to strike. We find that this tends to get caught by either our alignment audit or by the monitor substantially reducing in performance due to collusion.

2. Context-dependent misalignment: The model is narrowly trained to follow a specific malign strategy in exactly the situation we're worried about. We try to make it cover monitoring as well, but we find it hard to make it collude reliably enough, so the monitoring tends to flag a high enough rate of misbehavior that we are safe.

We try giving the red team different individual unfair advantages (e.g. visibly misaligned training data, training against monitors). In cases where the monitoring system uses the model of interest (e.g. for oversight), it is required to use the model organism. We try a number of approaches and experiment with interpolating between these two types of red-team models. We also experiment with varying details of the strategy, such as how often to strike and how carefully to try to maintain plausible deniability versus inflicting maximum damage. We also try to train steganographic reasoning to improve performance without increasing detection rates. Hopefully, none of our red-team models manage to defeat the safety measures.

Argument for safety

The argument here is similar to arguments made in the current report, but it focuses on a smaller set of crisp claims given the more robust safeguards and evaluations described above.

1. We think that causing a catastrophe via one of the threat pathways discussed requires planning and optimization for a bad outcome (even if just in a small number of isolated instances), rather than merely random mistakes.
 - a. *This would require some analysis to establish, and may not be true for e.g. leaking sensitive information.*
2. Thus, for a catastrophe to arise due to model misbehavior, the model has to actively pursue a harmful goal in some contexts. The model could do so by reasoning at deployment time about how to achieve the goal or by following heuristics that proved to be useful for tasks similar to the harmful task in training (or some combination of the two). The two types of red-team models above (consistent scheming and context-dependent misalignment) represent

two salient “model organisms” of plausible ways egregious misalignment could arise: either the model somehow acquired a malicious goal that survived training due to alignment faking, and/or it developed specific off-distribution malicious behaviors, possibly via unwanted generalization.

- a. The two archetypes are not mutually exclusive, but one key distinction is whether the optimization toward bad outcomes happens primarily via reinforcement during training or via planning during deployment.*
 - b. We’re not confident that this is the right way to think about egregious misalignment, but we think that creating red-team models that cover both archetypes should increase our confidence significantly.*
3. In “non-[concentrated](#)” (high-frequency) threat models: Sandbagging, poisoning model-generated training data.
 - a. We think that causing a catastrophe via these pathways requires malicious behavior in a large number of instances.
 - b. To pull off catastrophe, the model needs to misbehave often in the real settings, yet not misbehave at a detectable rate in our realistic, threat-relevant alignment audit settings.
 - i. Due to our monitoring, the model would need to distinguish these settings without verbalized reasoning, making its task even harder.
 - ii. We’re hoping: Despite some unfair advantages, our model organism efforts are not able to thread these needles at the model’s level of capability.
4. In “concentrated” (low-frequency) threat models: Sabotaging safety R&D, inserting backdoors, etc.
 - a. We think causing catastrophe via these pathways requires fairly blatant misbehavior, though there may still be plausible deniability in some cases.
 - i. To increase confidence in this claim, we train red-team models to validate that they still lack the capabilities to perform

catastrophically in subtle ways that would be non-obvious to a human even given assistance from a monitor.

- b. To pull off catastrophe, the model needs to not be caught by monitoring. Our monitoring is comprehensive, so this requires the monitor to either fail randomly or actively collude. However, we have tested that the monitor does in fact catch our test suite of sabotage attempts, so it would be unlikely for it to fail consistently enough to not flag anything across the actions required for catastrophe (which may be a small number but still enough to provide several opportunities for detection).
 - i. We're hoping: Despite some unfair advantages, our context-dependent misalignment model organism efforts are not able to thread these needles at the model's level of capability.
- c. Successful collusion could still be possible if the model is capable of making reliable careful distinctions between the honeypots and real failures. In particular, the distribution on which the model behaves catastrophically could be very narrow provided it occasionally arises in deployment. Being more consistently and coherently misaligned with more malicious reasoning at deployment time might increase the chance of success there, but would also increase the risk of detection by the alignment audit.
 - i. We're hoping: Despite some unfair advantages, our consistent scheming red-team models are not able to pass through our alignment audits.